第三讲(续) 变量重要性引申的问题

---预测模型解释中的SHAP方法

案例二：共享单车日租赁量的预测

- 案例二：共享单车日租赁量(Casual/Registered/count)的预测
- 案例二数据：

	season	holiday	workingday	weather	temp	atemp	humidity	windspeed	casual	registered	count
0	1	0	0	1	9.84	14.395	81	0.0	3	13	16
1	1	0	0	1	9.02	13.635	80	0.0	8	32	40
2	1	0	0	1	9.02	13.635	80	0.0	5	27	32
3	1	0	0	1	9.84	14.395	75	0.0	3	10	13
4	1	0	0	1	9.84	14.395	75	0.0	0	1	1

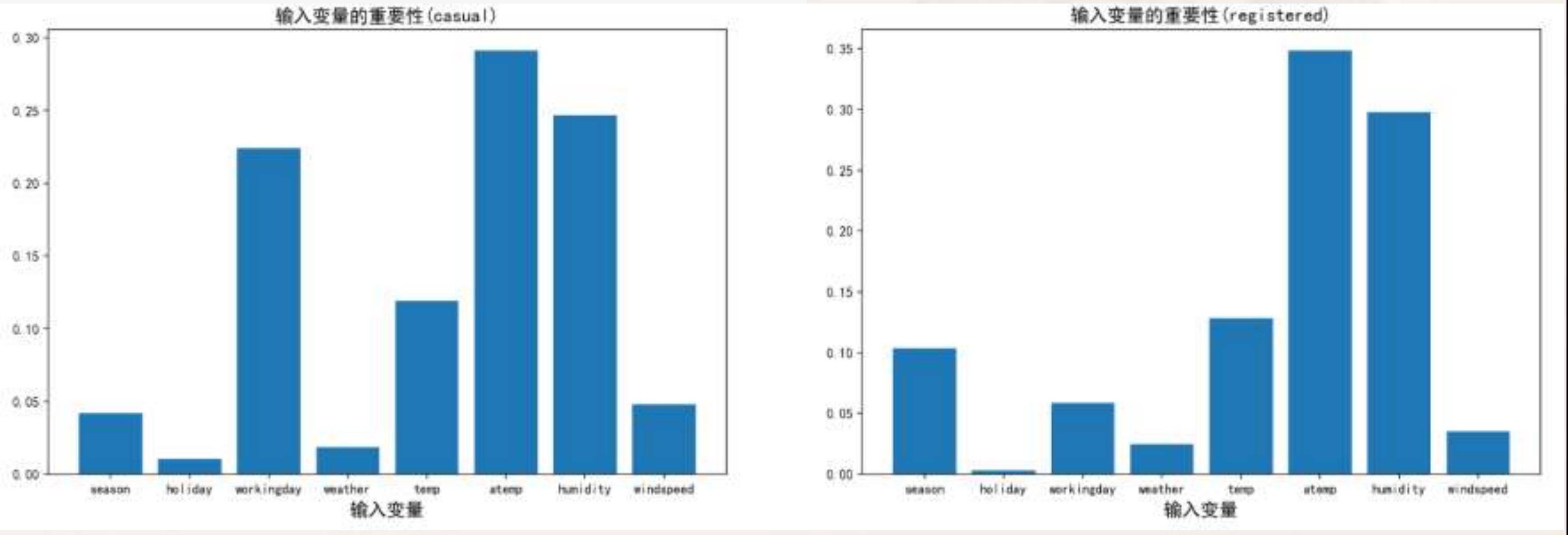
- Season – 1:spring 2:summer 3:fall 4:winter
- Holiday – whether the day is considered a holiday
- Workingday: whether the day is neither a weekend nor holiday
- Weather -
  - 1: Clear, Few clouds, Partly cloudy
  - 2: Mist + Cloudy, Mist + Broken clouds, Mist + Few clouds, Mist
  - 3: Light Snow, Light Rain + Thunderstorm + Scattered clouds, Light Rain + Scattered clouds
  - 4: Heavy Rain + Ice Pallets + Thunderstorm + Mist, Snow + Fog

- Temp - temperature in Celsius
- Atemp - "feels like" temperature in Celsius
- Humidity - relative humidity
- Windspeed - wind speed
- Casual – number of non-registered user rentals initiated
- Registered - number of registered user rentals initiated
- count - number of total rentals

变量重要性的再探讨

```
fig,axes = plt.subplots(nrows=1,ncols=2,figsize=(20,6),dpi=150)
for l,bestDeep,Y in zip([0,1],[9,7],['casual','registered']):
    y = data[Y]
    X_train, X_test, y_train, y_test = train_test_split(X,y,train_size = 0.90, random_state = 123)
    modelDTC = tree.DecisionTreeRegressor(max_depth = bestDeep,random_state = 123)
    modelDTC.fit(X_train,y_train)
    axes[l].bar(np.arange(8),modelDTC.feature_importances_)
    axes[l].set_title('输入变量的重要性(%s)'%(Y),fontsize=14)
    axes[l].set_xlabel('输入变量',fontsize=14)
    axes[l].set_xticks(np.arange(8))
    axes[l].set_xticklabels(X.columns,fontsize=10)
```

- 结论：影响两类骑行日租赁量的两大重要因素均为**体感温度和湿度**
- **问题与思考：**
  - 有哪些不足？



变量重要性的度量：Shapley值

- 联盟博弈论中一种评价项目中各联盟成员边际贡献的方法
  - Shapley值是对所有可能联盟下的所有边际贡献的平均值
    - 例：新房销售单价主要受五方面的影响：商业配套、产业、书包、医疗、地铁
      - 评估地铁对销售单价的边界贡献，所有联盟：
        - {Ø}± 地铁
        - {商业配套}±地铁、{产业}±地铁、{书包}±地铁、{医疗}±地铁
        - {商业配套+产业}±地铁、{商业配套+书包}±地铁、 {商业配套+医疗}±地铁、{产业+书包}±地铁、{产业+医疗}±地铁、{书包+医疗}±地铁
        - {商业配套+产业+书包}±地铁、{商业配套+产业+医疗}±地铁、{商业配套+书包+医疗}±地铁、{产业+书包+医疗}±地铁
        - {商业配套+产业+书包+医疗}±地铁

      - 地铁的Shapley值是上述所有联盟下地铁边界贡献的均值

- Shapley值体现了公平性，即贡献值反映了特征在所有可能的特征组合中的边际贡献

预测模型分析中的SHAP方法

- SHAP(SHapley Additive ExPlanations)
  - 提出了Shapley值计算和估计的不同方法(如TreeSHAP、KernelSHAP等)
  - 聚焦对个体预测的解释和变量重要性评估
- 利用Shapley值评价预测模型中输入变量x_j对输出变量预测值的边际贡献
  - 以线性回归预测模型为例： $\hat{f}(x)=\beta_0+\beta_1 x_1+\ldots+\beta_p x_p$
  - 任一样本观测$\boldsymbol{x}_i$为一个个体。x_j对$\boldsymbol{x}_i$预测值$\hat{f}(\boldsymbol{x}_i)$的边际贡献定义：

$$\phi_j^{(i)}=\beta_j x_{ij}-\mathrm{E}(\beta_j x_j)=\beta_j x_{ij}-\beta_j \mathrm{E}(x_j)$$

或正或负

  - 可基于$\phi_j^{(i)}$精确计算$\boldsymbol{x}_i$的预测值$\hat{f}(\boldsymbol{x}_i)$与平均预测值之差的成因

样本观测$\boldsymbol{x}_i$预测值与平均观测值的差

$$\hat{f}(\boldsymbol{x}_i)-\mathrm{E}\hat{f}(\boldsymbol{x})=\left(\beta_0+\sum_{j=1}^{p}\beta_j x_{ij}\right)-\left(\beta_0+\sum_{j=1}^{p}\mathrm{E}(\beta_j x_j)\right)=\sum_{j=1}^{p}\left(\beta_j x_{ij}-\mathrm{E}(\beta_j x_j)\right)=\sum_{j=1}^{p}\phi_j^{(i)}$$

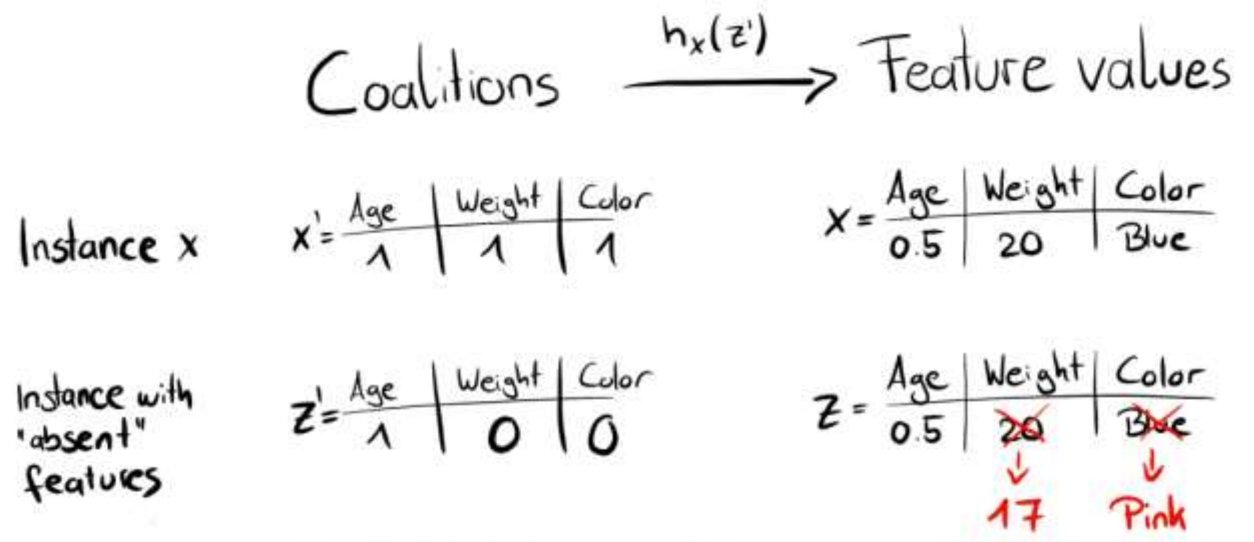
或正或负

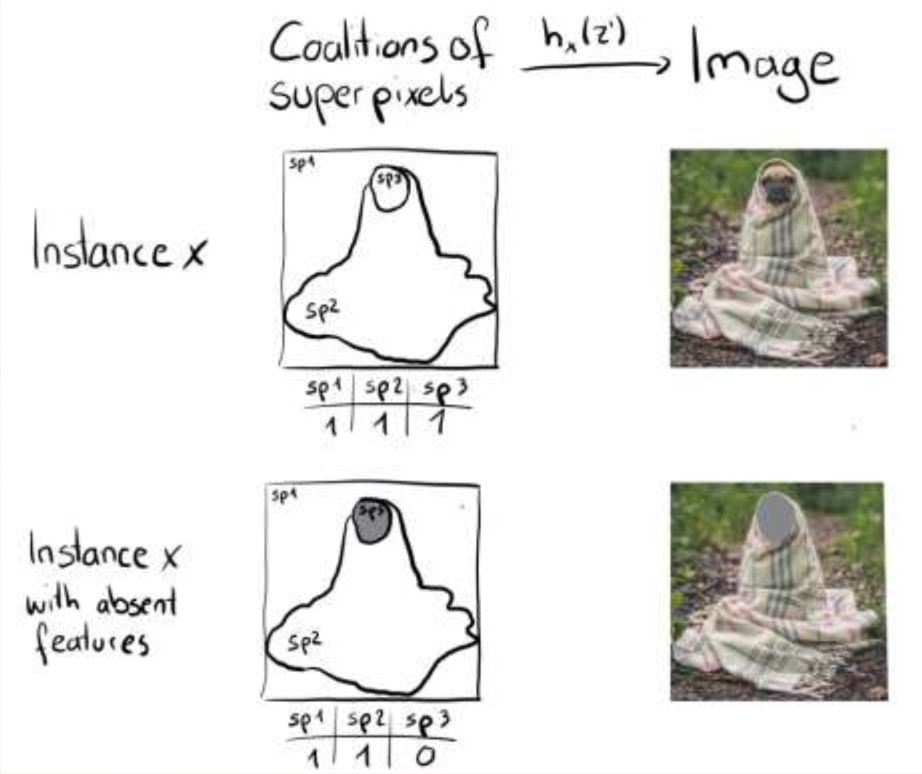
  - 输入变量x_j的整体重要性： $I_j=\sum_{i=1}^{n}|\phi_j^{(i)}|$

- $\boldsymbol{x}_i$的p个输入变量总的边际贡献(体现了Shapley值的可加性)
- 度量了在平均预测值基础上各输入变量的贡献"力"(拉高或拉低)→$\hat{f}(\boldsymbol{x}_i)$

估计Shapley值的基本思想

- 对任何类型的预测模型，如何计算输入变量x_j的Shapley值？
  - 对所有可能输入变量联盟，通过在模型中引入x_j和不引入x_j时预测值的差异来估计
  - 不在输入变量联盟中的变量值用随机值替换





- 对任何类型的预测模型，如何计算输入变量x_j的Shapley值？
  - 蒙特卡洛近似法：通过M次迭代计算输入变量x_j对样本观测$\boldsymbol{x}$的Shapley值
    - 对样本观测$\boldsymbol{x}$： $x = (x_{(1)}, \ldots, x_{(j)}, \ldots, x_{(p)})$
    - 从训练集中随机抽取样本观测$\boldsymbol{z_o}$： $z_o = (z_{(1)}, \ldots, z_{(j)}, \ldots, z_{(p)})$
    - 不在输入变量联盟中的变量值用随机值替换
      - 假设输入变量联盟中的变量有：$(x_1, x_2, \ldots x_{j-1})$

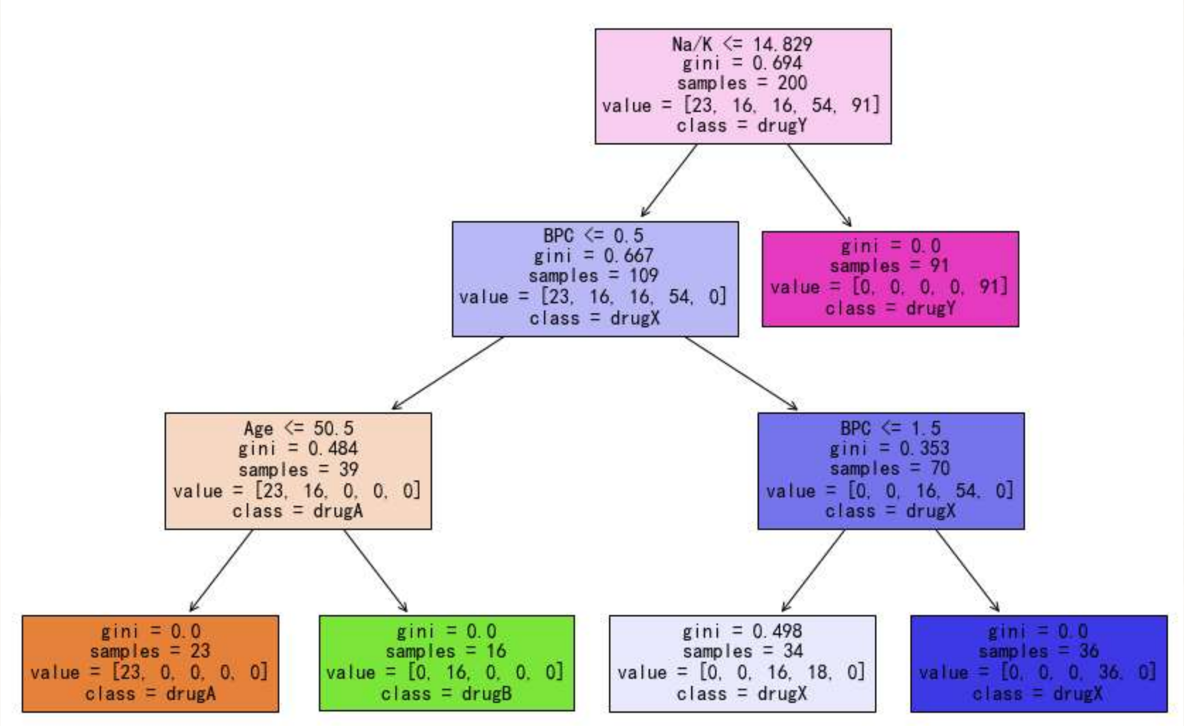
$x_{+j} = (x_{(1)}, \ldots, x_{(j-1)}, x_{(j)}, z_{(j+1)}, \ldots, z_{(p)})$ 引入x_j

$x_{-j} = (x_{(1)}, \ldots, x_{(j-1)}, z_{(j)}, z_{(j+1)}, \ldots, z_{(p)})$ 不引入x_j

$$\hat{\phi}_j = \frac{1}{M}\sum_{m=1}^{M}\left(\hat{f}(x_{+j}^m) - \hat{f}(x_{-j}^m)\right)$$

    - 需迭代M次以消除随机影响
  - 蒙特卡洛近似法的计算成本较高；M的取值没有很好的经验法则
  - 改进的算法
    - SHAP中的TreeSHAP、KernelSHAP等算法

- SHAP中的TreeSHAP：用于基于树模型(决策树、随机森林和梯度提升树)的机器学习
  - TreeSHAP速度快，可计算精确的Shapley值
  - 对于单棵树，利用树结构特点，根据每个节点上输入变量的分组，通过路径遍历计算输入变量贡献
  - 对于多棵树，对每棵树的Shapley值做加权平均即为最终结果
    - 例：



- 变量重要性的再探讨：基于决策树计算shapley值并分析影响日租赁量的主要因素

```
explainer = shap.TreeExplainer(modelDTC1)  #shap.Explainer(modelDTC)
shap_values = explainer(X)
print(shap_values)
```

针对causal的预测

```
X = data[['season','holiday','workingday','weather','temp','atemp','humidity','windspeed']]
```

各列依次对应输入变量

```
.values =
array([[ 3.88420197e+00,  1.28416457e-01,  1.28582438e+01, ...,
        -1.35564179e+01, -8.12505075e+00,  5.35726174e-01],
       [ 1.69732555e+00,  6.43004488e-02,  1.37912502e+01, ...,
        -1.62474275e+01, -1.03757833e+01, -1.81235243e+00],
       [-6.52629373e-01,  9.68160352e-01,  2.33340991e+01, ...,
        -1.70014932e+01, -4.44084793e+00,  2.79982113e+00],
       ...,
       [ 4.37664836e+00,  2.70345172e-02, -7.23222994e+00, ...,
        -1.30459696e+01,  9.07401835e-01, -1.47379768e-01],
       [ 3.69928395e+00,  6.78158245e-03, -9.35040217e+00, ...,
        -1.25546062e+01, -1.77037543e+00, -2.73657163e+00],
       [ 3.89092655e+00,  7.80558544e-03, -8.27991070e+00, ...,
        -1.29492974e+01, -6.04939794e+00, -8.06512446e-01]])
```

预测值的均值

```
.base_values =
array([36.62734148, 36.62734148, 36.62734148, ..., 36.62734148,
       36.62734148, 36.62734148])

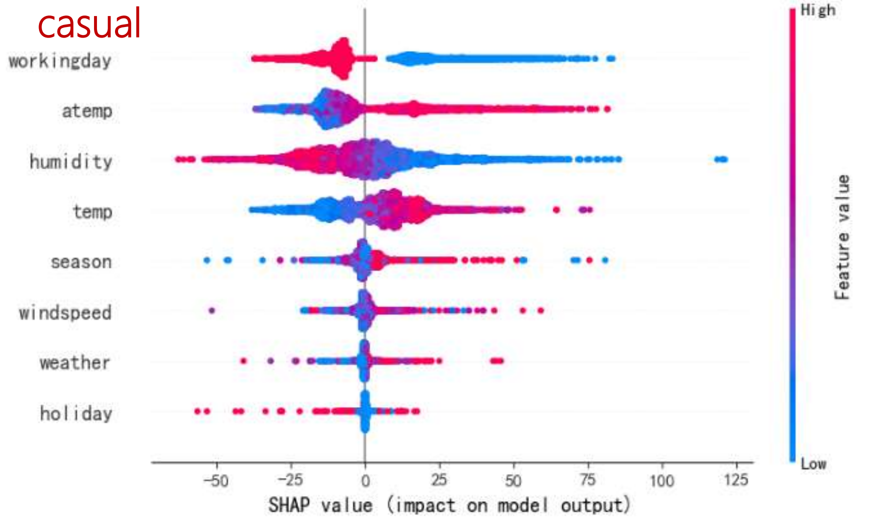
.data =
array([[ 1.     ,  0.     ,  0.     , ..., 12.88  , 75.     ,  6.0032],
       [ 1.     ,  0.     ,  0.     , ..., 19.695 , 76.     , 16.9979],
       [ 1.     ,  0.     ,  0.     , ..., 16.665 , 81.     , 19.0012],
       ...,
       [ 4.     ,  0.     ,  1.     , ..., 15.91  , 61.     , 15.0013],
       [ 4.     ,  0.     ,  1.     , ..., 17.425 , 61.     ,  6.0032],
       [ 4.     ,  0.     ,  1.     , ..., 16.665 , 66.     ,  8.9981]])
```

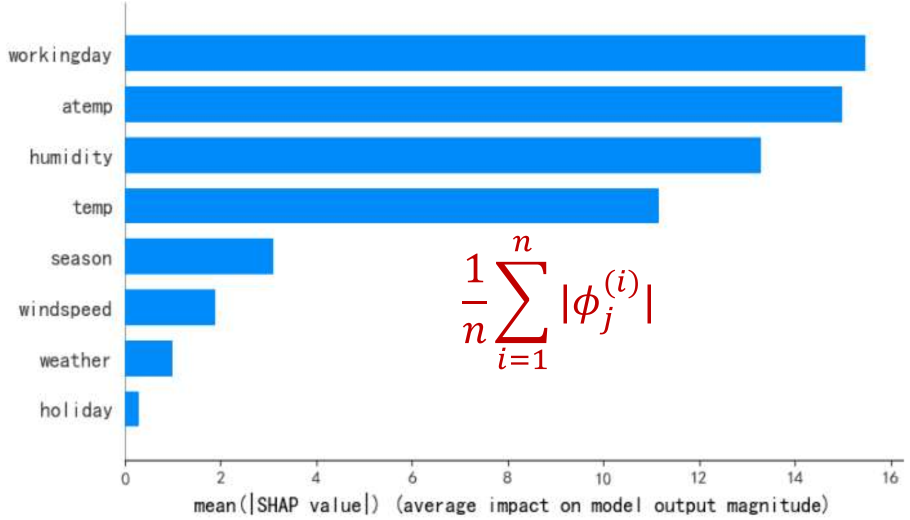
$$\phi_j^{(i)} = \beta_j x_{ij} - \mathrm{E}(\beta_j x_j) = \beta_j x_{ij} - \beta_j \mathrm{E}(x_j)$$

- 变量重要性的再探讨：基于决策树计算shapley值并分析影响日租赁量的主要因素
  - 利用SHAP**概要图**对影响因素进行可视化分析

```
shap.summary_plot(shap_values)
shap.summary_plot(shap_values,plot_type="bar")
```

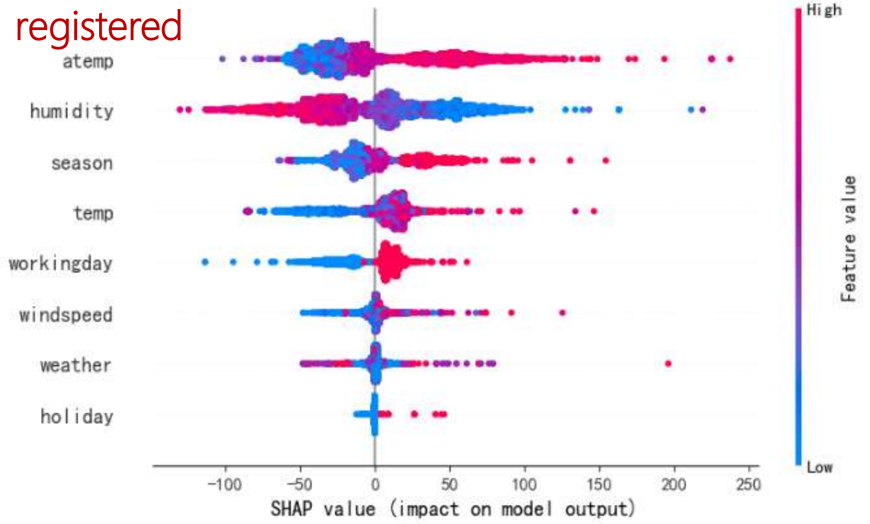
casual

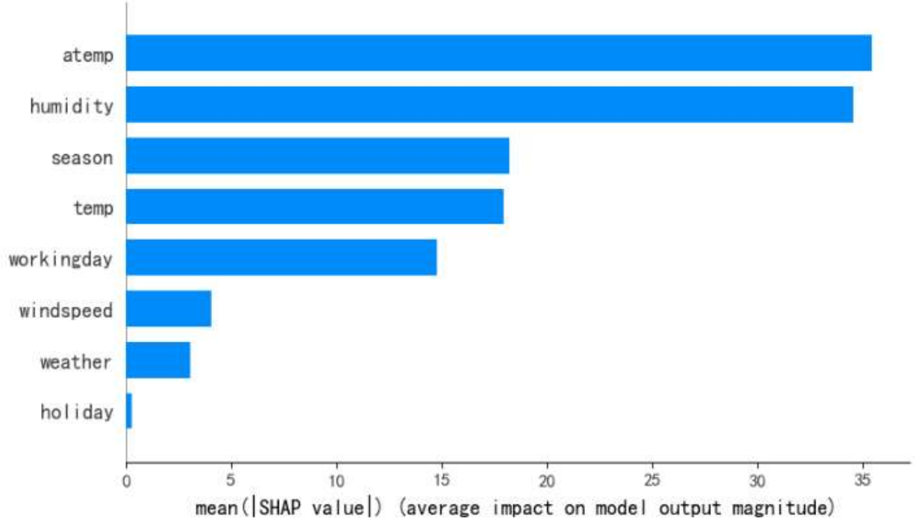




$$\frac{1}{n}\sum_{i=1}^{n}|\phi_j^{(i)}|$$

registered

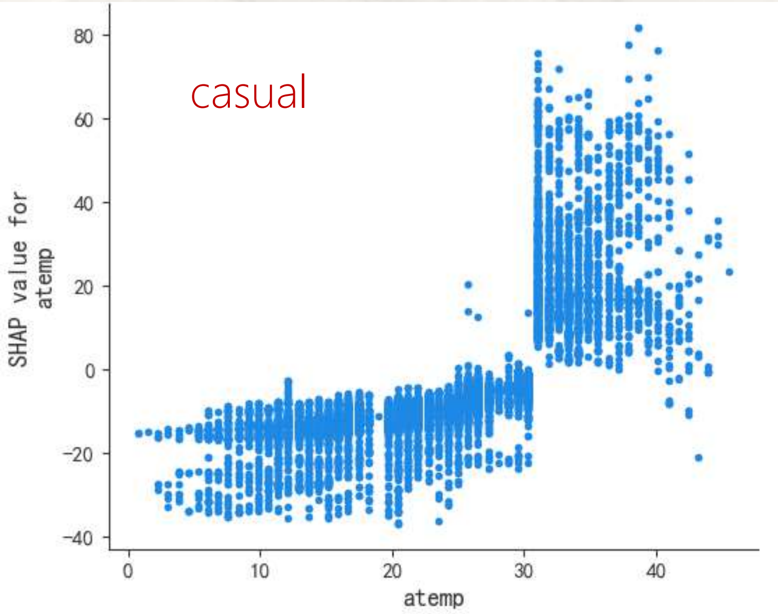


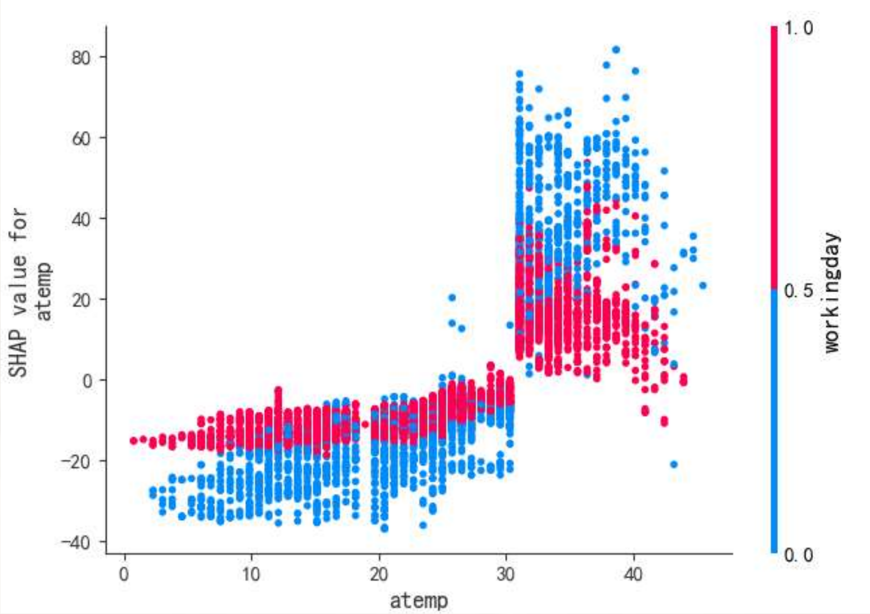


- 变量重要性的再探讨：基于决策树计算shapley值并分析影响日租赁量的主要因素
  - 利用依赖图和交互依赖图对重要因素的影响特点(强弱、线性或非线性)做可视化分析

```
shap.dependence_plot(ind = 5, shap_values =shap_values.values, features = X, feature_names = list(X.columns) , interaction_index = None)
shap.dependence_plot(ind = 5, shap_values =shap_values.values, features = X, feature_names = list(X.columns) , interaction_index = 2)
```

```
X = data[['season','holiday','workingday','weather','temp','atemp','humidity','windspeed']]
```





- 变量重要性的再探讨：基于决策树计算shapley值并分析影响日租赁量的主要因素
  - 利用力图对个体(单日)特征进行可视化分析

```
shap.force_plot(base_value=explainer.expected_value,shap_values=shap_values.values[1],feature_names=list(X.columns),
                features=shap_values.data[1],matplotlib=True)
```

$$\hat{f}(\boldsymbol{x}_i)-\mathrm{E}\hat{f}(\boldsymbol{x})=\sum_{j=1}^{p}\left(\beta_j x_{ij}-\mathrm{E}\left(\beta_j x_j\right)\right)=\sum_{j=1}^{p}\phi_j\left(\hat{f}(\boldsymbol{x}_i)\right)$$

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       [ 1.    ,  0.    ,  0.    , ..., 16.665 , 81.    , 19.0012],
       ...,
       [ 4.    ,  0.    ,  1.    , ..., 15.91  , 61.    , 15.0013],
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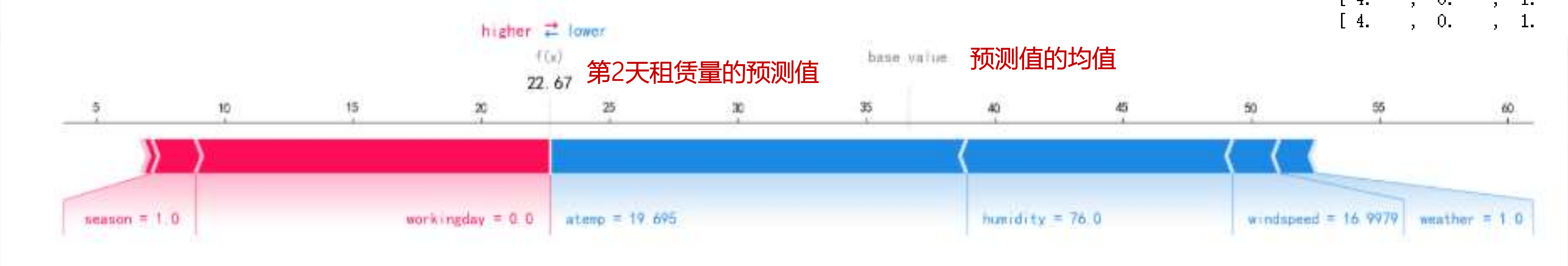
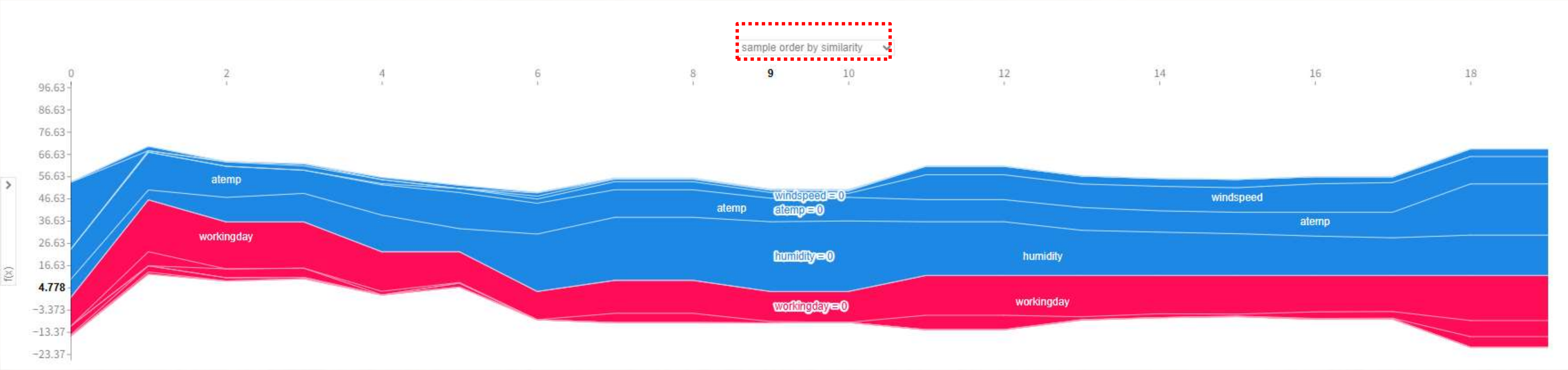
```



- 变量重要性的再探讨：基于决策树计算shapley值并分析影响日租赁量的主要因素
  - 利用聚类SHAP图对多个体特征进行可视化分析

```
shap.force_plot(base_value=explainer.expected_value,shap_values=shap_values.values[0:20],feature_names=list(X.columns))
```

聚类SHAP图：一组转置的力图



- Shapley值的推广应用
  - 基于Shapley值的聚类
  - 借助某分组(如季节，节假日等)力图分析各因素对分组预测值的影响，等

$$\hat{f}(\boldsymbol{x}_i) - \mathrm{E}\hat{f}(\boldsymbol{x}) = \sum_{j=1}^{p}\left(\beta_j x_{ij} - \mathrm{E}(\beta_j x_j)\right) = \sum_{j=1}^{p}\phi_j\left(\hat{f}(\boldsymbol{x}_i)\right)$$